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c. Microscopic slide

A slide of 60-years-old holotype specimen of a flatworm (*Lethacotyle fijiensis*) permanently mounted in canada balsam.



d. Ant inside blastic amber

Fossil resins-Amber

Plants secrete resins for their protective benefits.Amber is a fossilized tree resin especially from the wood, which has been appreciated for its colour and natural beauty since neolithic times. Much valued from antiquity to the present as a gemstone, amber is made into a variety of decorative objects. Amber is used in jewellery. It has also been used as a healing agent in folk medicine.

Figure 10.12: Economic importance of wood (a–d)

Secondary Phloem

The vascular cambial ring produces secondary phloem or bast on the outer side of the vascular bundle.

Just as the secondary xylem, the secondary phloem also has two tissue systems – the axial (vertical) and the radial (horizontal) systems derived respectively from the vertically elongated fusiform initials and horizontally elongated ray initials of vascular cambium. While sieve elements, phloem fibre, and phloem parenchyma represent the axial system, phloem rays represent the radial system. Life span of secondary phloem is less compared to secondary xylem. Secondary phloem is a living tissue that transports soluble organic compounds made during photosynthesis to various parts of plant.

Some commercially important phloem or bast fibres are obtained from the following plants.

i. Flax-*Linum ustitaissimum*

ii. Hemp-*Cannabis sativa*

iii. Sun hemp-*Crotalaria juncea*

iv. Jute-*Corchorus capsularis*

Be friendly with your environment (Eco friendly)

Why should not we use the natural products which are made by plant fibres like rope, fancy bags, mobile pouch, mat and gunny bags etc., instead of using plastics or nylon?

Periderm

Whenever stems and roots increase in thickness by secondary growth, the periderm, a protective tissue of secondary origin replaces the epidermis and often primary cortex. The periderm consists of phellem, phellogen, and phelloderm.

Phellem (Cork)

It is the protective tissue composed of non-living cells with suberized walls and formed centrifugally (outward) by the phellogen (cork cambium) as part of the periderm. It replaces the epidermis in